

Homework 3

1.

For context-free grammar G_4 ,

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

Give **BOTH** parse trees and derivations for each string:

- a. a
- b. $a \times a \times a + a$
- c. $a + a$
- d. $((a))$

2.

Answer each of the following questions for context-free grammar G .

$$\begin{aligned} R &\rightarrow XRX \mid S \\ S &\rightarrow aTb \mid bTa \\ T &\rightarrow XTX \mid X \mid \varepsilon \\ X &\rightarrow a \mid b \end{aligned}$$

- a. What are the variables of G ?
- b. What are the terminals of G ?
- c. Which is the start variable of G ?
- d. Give three strings in $L(G)$.
- e. True or False: $T \Rightarrow aba$.
- f. True or False: $T \overset{*}{\Rightarrow} aba$.
- g. True or False: $T \Rightarrow T$.
- h. True or False: $T \overset{*}{\Rightarrow} T$.
- i. True or False: $XXX \overset{*}{\Rightarrow} aba$.
- j. True or False: $X \overset{*}{\Rightarrow} aba$.
- k. True or False: $T \overset{*}{\Rightarrow} XX$.
- l. True or False: $T \overset{*}{\Rightarrow} XXX$.
- m. True or False: $S \overset{*}{\Rightarrow} \varepsilon$.